

Project Charters, Scope, Assumptions, and Constraints

DAEN 459 - Fall 2026
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- Explain what a project charter authorizes and what a scope statement bounds.
- Write measurable outcomes and explicit inclusions and exclusions.
- Separate assumptions (to validate) from constraints (to design within).
- Draft these statements for your own capstone using evidence and acceptance criteria.

One running example

Campus energy analytics: Facilities receives hourly meter exports in inconsistent CSV formats. Analysts spend hours cleaning them before reporting building electricity use.

- Proposed capstone: ingest historical and new exports into a documented, tested analytical dataset.
- Stakeholder: Facilities energy analyst; users: analysts preparing monthly reports.
- First decision: which buildings, fields, refresh frequency, and quality threshold matter?

Project charter: definition and purpose

A **project charter** is a short agreement that authorizes the project and records its purpose, sponsor, intended outcomes, boundaries at a high level, and basis for success.

- Problem and value: what decision or workflow will improve?
- Sponsor, users, team, and decision owner: who validates needs and accepts work?
- Objectives and deliverables: what evidence will show success?
- Time frame, major risks, dependencies, and approval: what makes the effort viable?

Charter statements: weak and strong

Weak

“Build a data pipeline for Facilities.”

“Improve data quality.”

“The team will make a dashboard.”

Strong

“By the capstone demo, deliver a scheduled pipeline for hourly electricity CSVs from 12 pilot buildings, with a documented table used by the Facilities analyst for monthly reporting.”

“For the pilot dataset, report completeness, duplicate meter-hours, and schema violations; achieve at least 98% valid records on the supplied acceptance sample.”

“Facilities analyst reviews a sample monthly usage query and signs off on the dataset and data dictionary in the final review.”

Scope definition: draw the boundary

The **scope statement** describes the work and outputs included in the project, what is excluded, and how completion will be accepted. It turns the charter's direction into a testable boundary.

- **In scope:** sources, time range, transformations, storage, interfaces, tests, documentation.
- **Out of scope:** tempting adjacent work and later phases.
- **Acceptance:** observable checks, reviewer, dataset, and conditions.
- **Change control:** record new requests and trade them against time and priorities.

Scope statements: weak and strong

Weak

“Integrate all energy data.”

“Create useful visualizations.”

“Pipeline is complete when it works.”

Strong

“Ingest hourly electricity CSV exports for 12 pilot buildings from January 2024 through June 2026; normalize meter IDs and timestamps into one curated table.”

“Deliver two reproducible SQL queries: monthly kWh by building and missing meter-hours by building; dashboard design is excluded.”

“On the acceptance sample, the scheduled run produces no duplicate meter-hour keys, at least 98% valid rows, and a run log; the Facilities analyst verifies both queries.”

Assumptions: claims to verify

An **assumption** is a condition treated as true for planning, although it has not yet been confirmed. Record its consequence, owner, and validation date.

- Data availability, access, schema stability, stakeholder availability, compute capacity.
- Test assumptions early with a sample, permission check, profiling, or stakeholder review.
- If an assumption fails, revise scope, method, or schedule explicitly.

Assumption statements: weak and strong

Weak

“The data is clean.”

“We can get access.”

“The API should be reliable.”

Strong

“We assume at least 95% of pilot files contain meter ID, timestamp, and kWh; the team will profile three months of files by week 2.”

“We assume Facilities can provide deidentified pilot exports by week 2; the sponsor will confirm access by Friday of week 1.”

“We assume the source posts a complete daily file by 06:00; compare seven arrival times and define a late-file alert by week 3.”

Constraints: limits to design within

A **constraint** is a known restriction on choices, resources, or delivery. State its source and practical implication.

- Examples: semester deadline, approved cloud budget, restricted student access, retention policy, mandated technology.
- Confirm whether a limit is truly fixed; a preferred tool is not necessarily a constraint.
- Design around constraints and note which deliverables they affect.

Constraint statements: weak and strong

Weak

“We have limited time.”

“Budget is small.”

“Data is private.”

Strong

“The final demo is due in week 14; ingestion and data quality checks must be ready for stakeholder review in week 11.”

“The team may use only the department’s existing storage and compute allocation; the design must not require a paid external service.”

“Student accounts may access only deidentified meter exports in the approved workspace; raw occupant or billing records are excluded.”

Keep the four parts consistent

1. **Charter:** why the pilot matters and who accepts it.
2. **Scope:** which 12 buildings, which records, which outputs, and which tests.
3. **Assumptions:** pilot exports arrive, contain required fields, and can be accessed.
4. **Constraints:** week 14 deadline, approved workspace, and available compute.

If an assumption fails, revisit the scope and the charter's success measure with the sponsor; document the decision.

Capstone drafting worksheet

Complete each sentence with your project's actual names, quantities, and dates.

1. **Charter:** For [user], [current problem] causes [impact]. By [date], we will deliver [outcome]; [sponsor] will accept it using [measure].
2. **Scope:** We include [sources, period, transformations, outputs, tests]. We exclude [adjacent work]. Completion means [test on named sample] reviewed by [person].
3. **Assumptions:** We assume [uncertain condition]. [Owner] will check it using [evidence] by [date]. If false, [response].
4. **Constraints:** [Source] limits us to [specific limit]; therefore [design or delivery implication].

Workshop/lab work: draft, challenge, revise

1. Draft one charter objective, one in-scope item, one exclusion, two assumptions, and two constraints for your capstone.
2. Pair review: underline every vague word (“all,” “easy,” “accurate,” “fast”); replace it with a boundary or measurable check.
3. Ask: who owns the data, what can fail first, what sample proves completion, and who can accept the result?
4. Revise the statements and log one unresolved decision for the sponsor.

Ready for sponsor review?

- Can a stakeholder explain the value and name the acceptance owner?
- Can another team identify exactly which data and outputs are included and excluded?
- Can each assumption be tested by a named person and date?
- Does each constraint have a source and an effect on the design?
- Is success demonstrable on a stated dataset or sample within the semester?